

Malware Analysis:

Malware, short for malicious software, refers to any program or file that is intentionally designed to harm a computer, network, or server. It can steal data, damage systems, or take control of your computer without your permission.

Types:

Viruses - A Virus is a malicious executable code attached to another executable file. The virus spreads when an infected file is passed from system to system.

Worms - Worms replicate themselves on the system, attaching themselves to different files and looking for pathways between computers, such as computer network that shares common file storage areas. Worms usually slow down networks. A virus needs a host program to run but worms can run by themselves.

Trojan horse - A [Trojan horse](#) is malware that carries out malicious operations under the appearance of a desired operation such as playing an online game.

Ransomware - [Ransomware](#) encrypts data in the computer with a key that is unknown to the user. The user has to pay a ransom (price) to the criminals to retrieve data.

Adware - It displays unwanted ads and pop-ups on the computer.

Spyware - Its purpose is to steal private information from a computer system for a third party.

Rootkits - A [rootkit](#) modifies the OS to make a backdoor. Attackers then use the backdoor to access the computer distantly.

Keyloggers - Keylogger records everything the user types on his/her computer system to obtain passwords and other sensitive information and send them to the source of the keylogging program.

What is Malware Analysis?

Malware Analysis is the process of examining a suspicious file or URL to determine its functionality, origin and potential impact.

This involves understanding how malware works and identifying its behavior.

Types of Malware Analysis:

- Static Analysis
 - Analyze malware without executing it
- Dynamic Analysis
 - Analyze malware by executing its source code

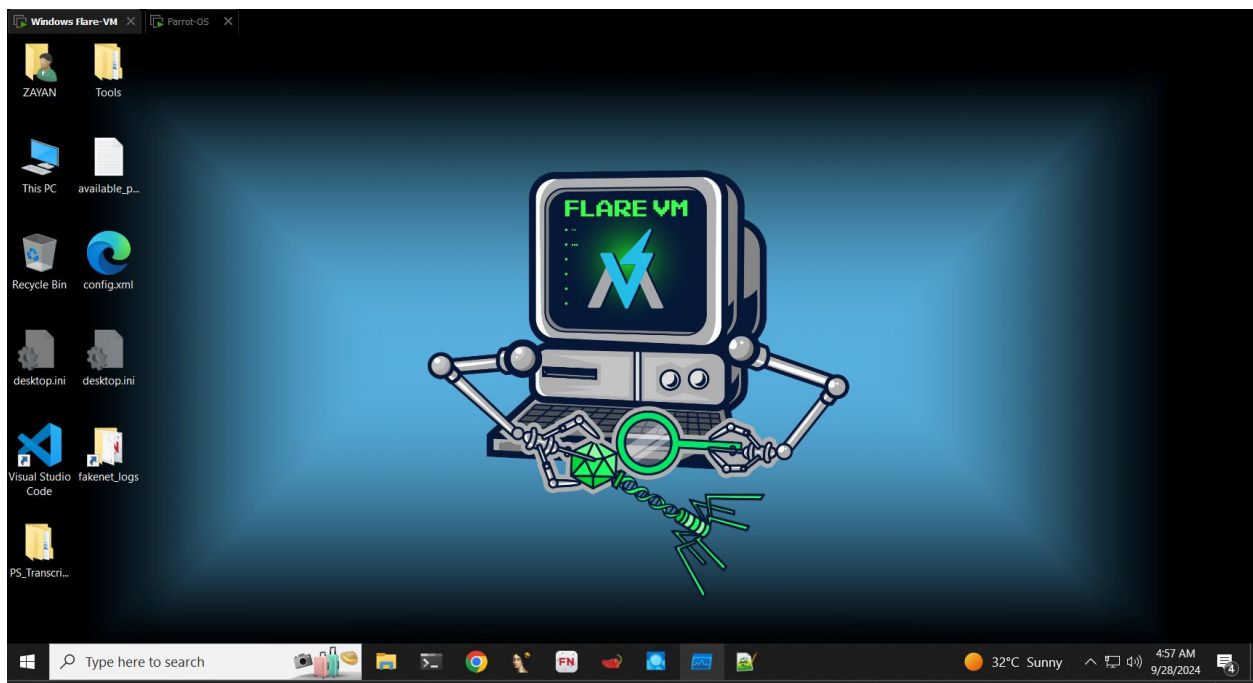
Lab: Buildup of Malware Analysis Environment

- Install “**FlareVM**” on Windows 10 virtual machine.
- Download “**PMAT**” malware samples from [github](#).
- Install “**INetSim**” on [ParrotOS](#) and prepare Flare VM to use [ParrotMachine](#) (with [INetSim](#) settings) for fake Internet simulation. Verify
- Safely copy malware sample (**Rams.exe**) from base machine to Analysis virtual machine.
- Secure / Isolate your malware analysis environment from accidental breakout.

Installation of Flare-VM:

1. Download Flare-VM from Github > Extract it > Run powershell as an administrator > Write this command **Set-ExecutionPolicy Unrestricted**
2. Navigate to the folder where flare-vm is **cd .. > cd .. > cd ZAYAN > cd Downloads > cd flare-vm main > .\install.ps1**

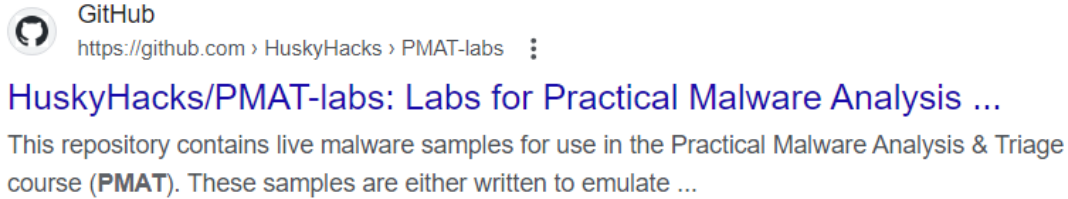
It will start installing VM.



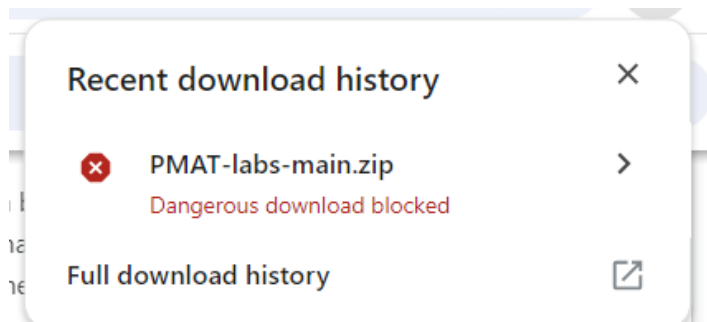
It will look like this after installation.

Installation of PMAT Labs:

1. Search for pmat labs



2. Search on github pmat –labs (Husky hacks)> click on souce code .zip > download suspicious file from the history > extract >



3. Inside Pmat there is a folder of pmat labs > clicks on labs to see its content.

Remember: Do not open any file

Static Malware Analysis:

Lab: Static Malware Analysis

- Perform static malware analysis on provided sample (**Rams.exe**) as under:-
 - Find Portable Executable (PE) information of malware executable file using “**PEstudio**” to determine following information:-
 - SHA-256 Hash value
 - File type. Note the values for "first-byte-hex" and "first-bytes-text" for later verification/analysis
 - **VirusTotal** Detection result
 - Confirm above information via “**CFF explorer**” and “**Online TrID File Identifier**”
 - Analyze Executable file and carryout fingerprinting
 - Strings search using “**Floss**”, “**BinText**” and “**Strings**” tool
 - Fuzzy hashing using “**ssdeep**”
 - Packaging and Obfuscation:
 - Use “**UPX**” tool for packing a malware sample “**TotalAware3.exe**”. Also carry out unpacking of sample using UPX.
 - Compare packed and unpacked version of the malware using CFF explorer.
 - Verify the type of malware analyzing using CFF (Import Directory).
 - Repeat the confirmation of obfuscation using “**IDA Pro**”

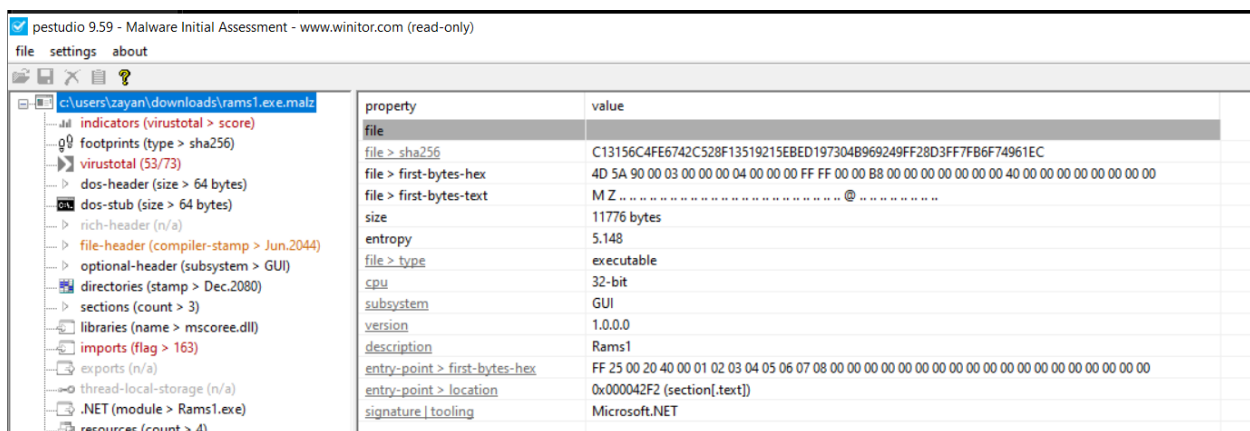
In Flare-VM > Tools > PE > pestudio > Run as an administrator

Put malware in this tool.

If we put malware in **pestudio**, it will extract information from it like type of hash, first byte hex, first byte text etc.

First byte hex will help us to identify which type of file it is. 4D 5A means it is .exe file

Look for **entropy** value to determine severity and randomness of this malware.



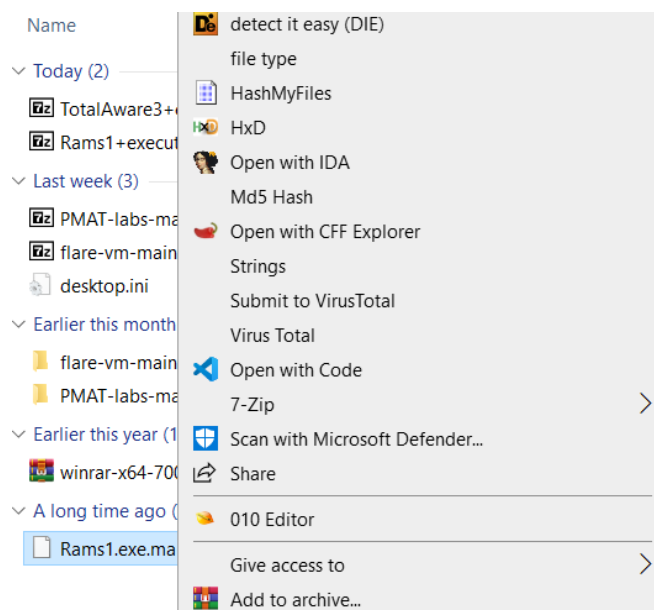
The screenshot shows the PESTudio 9.59 interface. The left pane displays a tree view of file properties for 'c:\users\zayan\downloads\rams1.exe.malz'. The right pane shows a table of properties and their values.

property	value
file	
file > sha256	C13156C4FE6742C528F13519215EBED197304B969249FF28D3FF7FB6F74961EC
file > first-bytes-hex	4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
file > first-bytes-text	MZ
size	11776 bytes
entropy	5.148
file > type	executable
cpu	32-bit
subsystem	GUI
version	1.0.0.0
description	Rams1
entry-point > first-bytes-hex	FF 25 00 20 40 00 01 02 03 04 05 06 07 08 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
entry-point > location	0x000042F2 (section[.text])
signature tooling	Microsoft.NET

indicators (virustotal > score)	ALYac	-	30.08.2024	31
footprints (type > sha256)	APEX	-	28.08.2024	33
virustotal (53/73)	AVG	Win32:Trojan-gen	30.08.2024	31
dos-header (size > 64 bytes)	Acronis	-	28.03.2024	186
dos-stub (size > 64 bytes)	AhnLab-V3	Trojan/Win.Generic.C4403926	30.08.2024	31
rich-header (n/a)	Alibaba	Ransom:MSIL/Ryzerlo.830e8c1d	27.05.2019	1953
file-header (compiler-stamp > Jun.2044)	Arcabit	Trojan.Ransom.RTH.1	29.08.2024	32
optional-header (subsystem > GUI)	Avast	Win32:Trojan-gen	30.08.2024	31
directories (stamp > Dec.2080)	Avira	HEUR/AGEN.1309291	30.08.2024	31
sections (count > 3)	Baidu	-	18.03.2019	2023
libraries (name > mscoree.dll)	BitDefender	Gen:Heur.Ransom.RTH.1	30.08.2024	31
imports (flag > 163)	BitDefenderTheta	Gen:NN.ZemsiF.36812.am0@aCaQ!Yi	16.08.2024	45
exports (n/a)	Bkav	W32.AIDetectMalware.CS	29.08.2024	32
thread-local-storage (n/a)	CAT-QuickHeal	-	29.08.2024	32
.NET (module > Rams1.exe)	CMC	-	29.08.2024	32
resources (count > 4)	ClamAV	-	29.08.2024	32
strings (flag > 2)	CrowdStrike	win/malicious_confidence_100% (W)	26.10.2023	340
debug (stamp > Dec.2080)	Cybereason	malicious.c561f4	18.08.2024	43
manifest (level > asInvoker)	Cylance	Unsafe	29.08.2024	32
version (FileDescription > Rams1)	Cynet	-	29.08.2024	32
certificate (n/a)	DeepInstinct	MALICIOUS	31.07.2024	61
overlay (n/a)	DrWeb	Trojan.EncoderNET.95	30.08.2024	31
	ESET-NOD32	a variant of MSIL/Filecoder.AK	30.08.2024	31
	Elastic	malicious (high confidence)	02.08.2024	59
	Emsisoft	Gen:Heur.Ransom.RTH.1 (B)	30.08.2024	31
	F-Secure	Heuristic.HEUR/AGEN.1309291	29.08.2024	32
	FireEye	Generic.mg.84f457ec561f492f	29.08.2024	32
	Fortinet	MSIL/Filecoder.AK!tr.ransom	30.08.2024	31
	GData	MSIL.Trojan-Ransom.Cryptear.X	29.08.2024	32

- **Virus-total** telling us that 53 out of 73 components tell us that it's a malware.

In Flare-VM, right click on malware file and in drop-down click **HashmyFile** to find about different hashes of this malware.



HashMyFiles					
File Edit View Options Help					
Filename	MD5	SHA1	CRC32	SHA-256	SHA-512
Rams1.exe.malz	84f457ec561f492f487d82eba049fd74	9e9294dad441369f953d2f97a9fd6c76b258e0...	5c61d061	c13156c4fe6742c528f13519215ebd197304b...	2662f5f5404a61...

You can also check it with **CFF Explorer**

CFF Explorer VIII - [Rams1.exe.malz]

File Settings ?

Rams1.exe.malz

Property	Value
File Name	C:\Users\ZAYAN\Downloads\Rams1.exe.malz
File Type	Portable Executable 32 .NET Assembly
File Info	No match found.
File Size	11.50 KB (11776 bytes)
PE Size	11.50 KB (11776 bytes)
Created	Tuesday 11 August 2020, 19.11.42
Modified	Friday 01 May 2020, 14.50.15
Accessed	Monday 30 September 2024, 08.17.33
MD5	84F457EC561F492F487D82EBA049FD74
SHA-1	9E9294DAD441369F953D2F97A9FD6C76B258E030

File: Rams1.exe.malz

- Dos Header
- Nt Headers
 - File Header
 - Optional Header
 - Data Directories [x]
- Section Headers [x]
- Import Directory
- Resource Directory
- Relocation Directory
- Debug Directory
- .NET Directory
 - MetaData Header
 - MetaData Streams
- #~

CFF Explorer VIII - [Rams1.exe.malz]

File Settings ?

Rams1.exe.malz

VA	RVA	File Offset

File: Rams1.exe.malz

- Dos Header
- Nt Headers
 - File Header
 - Optional Header
 - Data Directories [x]
- Section Headers [x]
- Import Directory
- Resource Directory
- Relocation Directory
- Debug Directory
- .NET Directory
 - MetaData Header
 - MetaData Streams
- #~
 - Tables Header
 - Tables
 - #Strings
 - #US
 - #GUID
 - #Blob
- Address Converter
- Dependency Walker
- Hex Editor
- Identifier
- Import Adder
- Quick Disassembler
- Rebuilder
- Resource Editor

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	Ascii
00000000	4D	5A	90	00	03	00	00	04	00	00	00	FF	FF	00	00	00	MZ.....yy..
00000010	B8	00	00	00	00	00	00	40	00	00	00	00	00	00	00	00	@.....
00000020	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	@.....
00000030	00	00	00	00	00	00	00	00	00	00	00	00	80	00	00	00	!.....
00000040	0E	1F	BA	0E	00	B4	09	CD	21	B8	01	4C	CD	21	54	68	0 00 .I.I.II.Th
00000050	69	73	20	70	72	6F	67	72	61	6D	20	63	61	6E	6E	6F	is.program.canno
00000060	74	20	62	65	20	72	75	6E	20	69	6E	20	44	4F	53	20	t.be.run.in.DOS.
00000070	6D	6F	64	65	2E	0D	0D	0A	24	00	00	00	00	00	00	00	mode...\$.
00000080	5D	45	00	00	4C	01	03	00	F2	2D	F9	8B	00	00	00	00	PE..I..0-0I....
00000090	00	00	00	00	E0	00	22	00	0B	01	30	00	00	24	00	00	...A".000\$.
000000A0	00	08	00	00	00	00	00	00	F2	42	00	00	00	20	00	00	..0...0B.....
000000B0	00	60	00	00	00	40	00	00	20	00	00	00	00	02	00	00	@B.....
000000C0	04	00	00	00	00	00	00	06	00	00	00	00	00	00	00	00	@.....
000000D0	00	A0	00	00	02	00	00	00	00	00	00	02	00	60	85	00	@.....
000000E0	00	00	10	00	00	00	00	20	00	10	00	00	00	00	00	00	@.....
000000F0	00	00	00	00	10	00	00	00	00	00	00	00	00	00	00	00	@.....
00000100	9D	42	00	00	4F	00	00	00	00	60	00	00	8C	05	00	00	B..O.....0..
00000110	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	@B.....
00000120	00	80	00	00	0C	00	00	00	24	42	00	00	38	00	00	00	..I..I..\$B..8..
00000130	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	@.....
00000140	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	@.....
00000150	00	00	00	00	00	00	00	00	20	00	00	00	08	00	00	00	@.....
00000160	00	00	00	00	00	00	00	08	20	00	00	48	00	00	00	00	@.....
00000170	00	00	00	00	00	00	00	2E	74	65	78	74	00	00	00	00	@.....
00000180	00	23	00	00	00	20	00	00	00	24	00	00	02	00	00	00	..#.....\$.
00000190	00	00	00	00	00	00	00	00	00	00	00	20	00	00	60	00	@.....
000001A0	2E	72	73	72	63	00	00	8C	05	00	00	00	60	00	00	00	..rsrc..@.....
000001B0	00	06	00	00	26	00	00	00	00	00	00	00	00	00	00	00	@.....
000001C0	00	00	00	40	00	40	2E	72	65	6C	6F	63	00	00	00	00	@.....
000001D0	0C	00	00	00	00	80	00	00	02	00	00	2C	00	00	00	00	@.....
000001E0	00	00	00	00	00	00	00	00	00	00	40	00	00	42	00	00	@.....

We have used multiple tools inside Flare-VM for surety of information because it is not best practice to rely on just 1 tool.

C > Program Data > Chocolatey > bin (Here you can find all .exe files)

Open terminal here in **Flare-VM FLOSS.exe** followed by malware name.

e.g: FLOSS.exe Ramz.exe

```
FLARE-VM Mon 09/30/2024 8:26:40.11
C:\Users\ZAYAN\Downloads>FLOSS.exe Ramz.exe.malz|
```

This tool extracts all the strings from malware. **Strings** are text or characters inside malware code. They help us to identify what this malware is doing whether it is creating backdoor to some site or which commands this malware is executing.

```
-----
FLOSS STATIC STRINGS (284)
-----
+-----+
| FLOSS STATIC STRINGS: ASCII (217) |
+-----+

!This program cannot be run in DOS mode.
.text
.rsrc
@.reloc
BSJB
v4.0.30319
#Strings
#GUID
#Blob
IEnumerable`1
Form1
Rams1
66840DDA154E8A113C31DD0AD32F7F3A366A80E8136979D8F5A101D3D29D6F72
SHA256
get_UTF8
<Module>
<PrivateImplementationDetails>
SizeF
System.IO
set_IV
mscorlib
System.Collections.Generic
Form1_Load
add_Load
RijndaelManaged
```



```

+-----+
| FLOSS STATIC STRINGS: UTF-16LE (67) |
+-----+

C:\Users\
\test\
http://alt356.onion/keys/save.php
Your files have been encrypted with Rams1, your id number is
, follow these instructions: http://bercerramsh.onion/2834-234323-23490-54665
\Desktop\ReadmeForDecryption.txt
<>/. , :-_+=}{[]()&*#@!ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890abcdefghijklmnopqrstuvwxyz
.iso
.bat
.txt

```

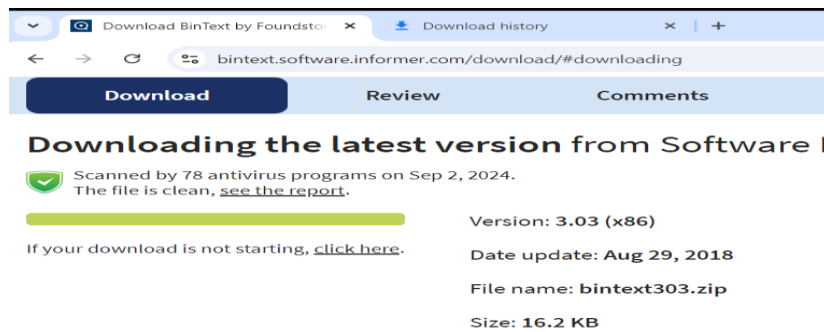
If we read this, we can see it is redirecting our machine to some onion site and also telling us our files have been encrypted. From this, we know this is a ransomware.

Strings extracted from pestudio:

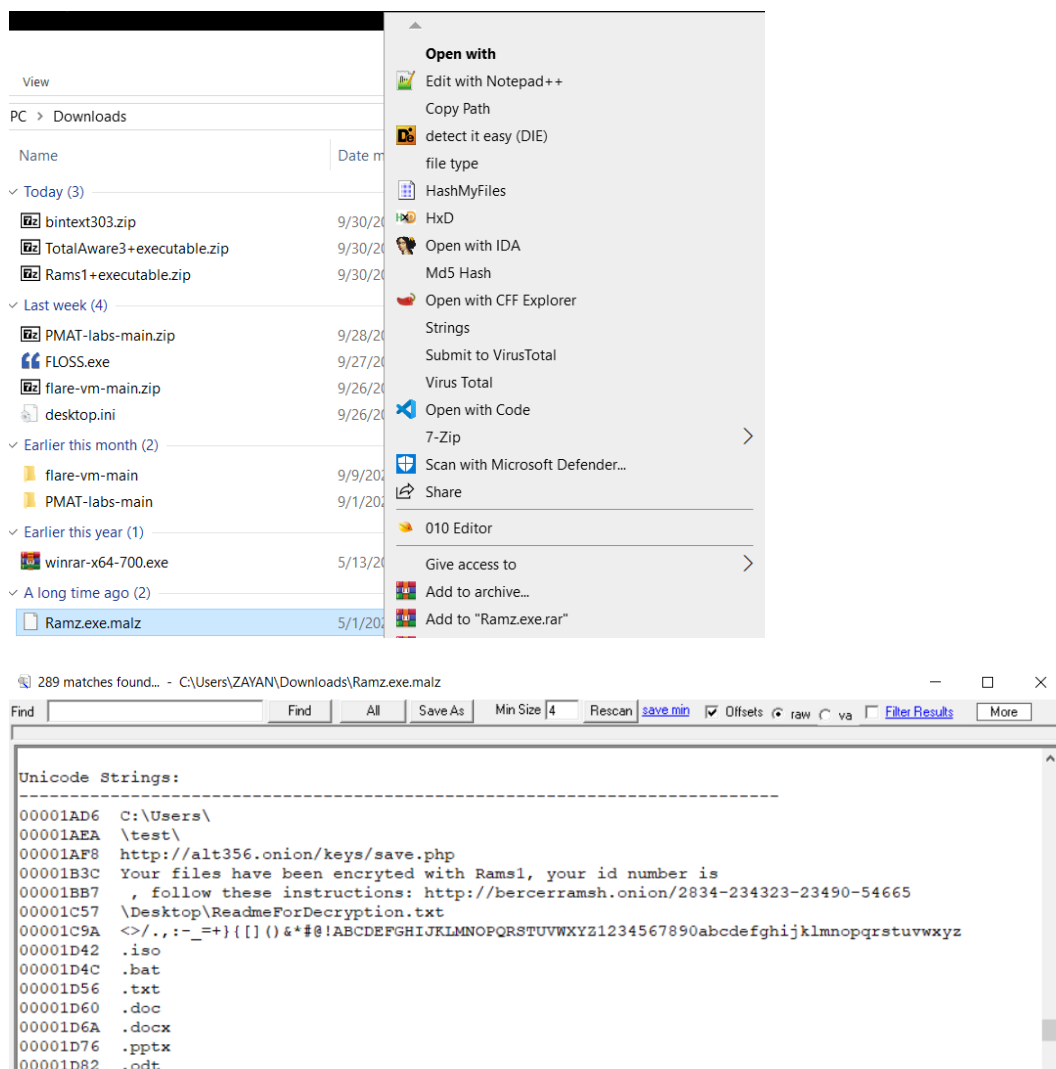
pestudio 9.59 - Malware Initial Assessment - www.winitor.com (read-only)

	encoding (2)	size (bytes)	location	flag (2)	label (321)	group (4)	value (508)
indicators (virusotal > score)	ascii	15	section:text	-	import	obfuscation	CreateEncryptor
footprints (type > sha256)	ascii	15	section:text	-	import	obfuscation	CreateEncryptor
virusotal (33/73)	ascii	12	section:text	x	import	memory	MemoryStream
dos-header (size > 64 bytes)	ascii	12	section:text	x	import	memory	MemoryStream
dos-stub (size > 64 bytes)	ascii	13	section:text	-	import	file	WriteAllBytes
rich-header (n/a)	ascii	12	section:text	-	import	file	WriteAllText
file-header (compiler-stamp > Jun.2044)	ascii	13	section:text	-	import	file	WriteAllBytes
optional-header (subsystem > GUI)	ascii	12	section:text	-	import	file	WriteAllText
directories (stamp > Dec.2080)	ascii	15	section:text	-	import	crypto	RijndaelManaged
sections (count > 3)	ascii	16	section:text	-	import	crypto	CryptoStreamMode
libraries (name > mscoree.dll)	ascii	10	section:text	-	import	crypto	CipherMode
imports (flag > 163)	ascii	12	section:text	-	import	crypto	CryptoStream
exports (n/a)	ascii	16	section:text	-	import	crypto	ICryptoTransform
thread-local-storage (n/a)	ascii	24	section:text	-	import	crypto	RNGCryptoServiceProvider
.NET (module > Rams1.exe)	ascii	15	section:text	-	import	crypto	RijndaelManaged
resources (count > 4)	ascii	16	section:text	-	import	crypto	CryptoStreamMode
strings (flag > 2)	ascii	10	section:text	-	import	crypto	CipherMode
debug (stamp > Dec.2080)	ascii	12	section:text	-	import	crypto	CryptoStream
manifest (level > asinvoker)	ascii	16	section:text	-	import	crypto	ICryptoTransform
version (FileDescription > Rams1)	ascii	24	section:text	-	import	crypto	RNGCryptoServiceProvider
certificate (n/a)	ascii	64	section:text	-	size	-	66840DDA154E8A113C31DD0AD32F7F3A366A80E8136979D8F5A101D3C
overlay (n/a)	ascii	40	dos-stub	-	dos-message	-	!This program cannot be run in DOS mode.
	ascii	13	section:text	-	import	-	IEnumerable`1
	ascii	64	section:text	-	size	-	66840DDA154E8A113C31DD0AD32F7F3A366A80E8136979D8F5A101D3C
	ascii	8	section:text	-	import	-	get_UTF8
	ascii	8	section:text	-	import	-	<Module>
	ascii	30	section:text	-	import	-	<PrivateImplementationDetails>
	ascii	8	section:text	-	import	-	mscorlib

- Download **bintext** tool. It is also for extracting strings. It is GUI based tool.



We can also find strings using built-in tool in Flare-VM. Just right click on sample and click on strings.



Packing of sample using **UPX Tool**:

- Open terminal in Flare-VM > upx.exe (It will open help section like what you can do with this tool)

For packing, use the following command:

- **upx.exe TotalAware3.exe**

For unpacking, use this:

- **upx.exe -d TotalAware3.exe**

```
FLARE-VM Mon 09/30/2024 8:44:16.45
C:\Users\ZAYAN\Downloads>upx.exe TotalAware3.exe
                        Ultimate Packer for eXecutables
                        Copyright (C) 1996 - 2024
UPX 4.2.4             Markus Oberhumer, Laszlo Molnar & John Reiser

  File size      Ratio      Format      Name
  -----
  111104 ->    26624    23.96%    win32/pe    TotalAware3.exe

Packed 1 file.
```

```
FLARE-VM Mon 09/30/2024 8:44:42.56
C:\Users\ZAYAN\Downloads>upx.exe -d TotalAware3.exe
                        Ultimate Packer for eXecutables
                        Copyright (C) 1996 - 2024
UPX 4.2.4             Markus Oberhumer, Laszlo Molnar & John Reiser

  File size      Ratio      Format      Name
  -----
  111104 <-    26624    23.96%    win32/pe    TotalAware3.exe

Unpacked 1 file.
```

- Open this sample with **CFF Explorer**, you'll find info like UPX. This means this file is packed.

CFF Explorer VIII - [TotalAware3.exe]

File Settings ?

TotalAware3.exe

Member	Offset	Size	Value	Section
Export Directory RVA	00000158	Dword	00000000	
Export Directory Size	0000015C	Dword	00000000	
Import Directory RVA	00000160	Dword	000331DC	UPX0
Import Directory Size	00000164	Dword	00000164	
Resource Directory RVA	00000168	Dword	00033000	UPX0
Resource Directory Size	0000016C	Dword	000001DC	
Exception Directory RVA	00000170	Dword	00000000	
Exception Directory Size	00000174	Dword	00000000	
Security Directory RVA	00000178	Dword	00000000	
Security Directory Size	0000017C	Dword	00000000	
Relocation Directory RVA	00000180	Dword	00033340	UPX0
Relocation Directory Size	00000184	Dword	00000010	
Debug Directory RVA	00000188	Dword	00000000	
Debug Directory Size	0000018C	Dword	00000000	
Architecture Directory RVA	00000190	Dword	00000000	
Architecture Directory Size	00000194	Dword	00000000	
Reserved	00000198	Dword	00000000	
Reserved	0000019C	Dword	00000000	
TLS Directory RVA	000001A0	Dword	00000000	
TLS Directory Size	000001A4	Dword	00000000	
Configuration Directory RVA	000001A8	Dword	00032E24	UPX0
Configuration Directory Size	000001AC	Dword	000000B8	
Bound Import Directory RVA	000001B0	Dword	00000000	

Also, the kernel processes are more like 23 or 24 but it is showing only 4. This is also a sign that this is a packed file.

CFF Explorer VIII - [TotalAware3.exe]

File Settings ?

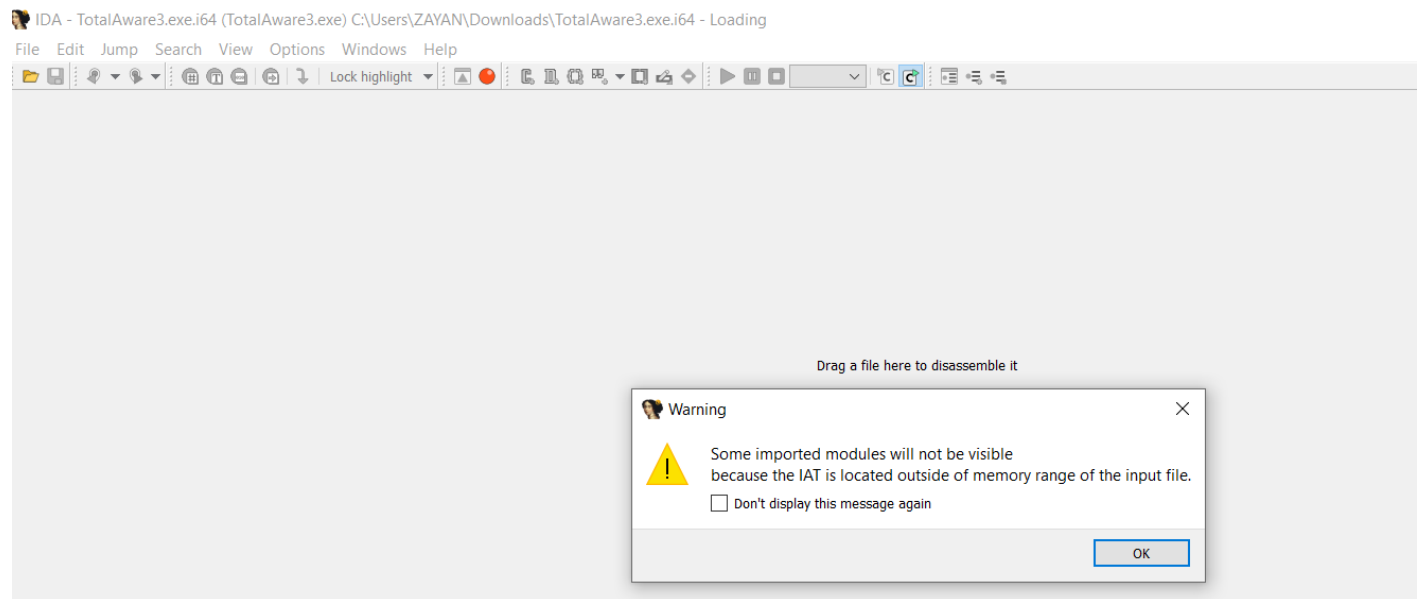
TotalAware3.exe

Module Name	Imports	OFTs	TimeDateStamp	ForwarderChain	Name RVA	FTs (IAT)
00006688	N/A	000065DC	000065E0	000065E4	000065E8	000065EC
szAnsi	(nFunctions)	Dword	Dword	Dword	Dword	Dword
KERNEL32.DLL	4	00000000	00000000	00000000	00033288	00033254
MSVCP140D.dll	1	00000000	00000000	00000000	00033295	00033268
ucrtbased.dll	1	00000000	00000000	00000000	000332A3	00033270
USER32.dll	1	00000000	00000000	00000000	000332B1	00033278
VCRUNTIME140D.dll	1	00000000	00000000	00000000	000332BC	00033280

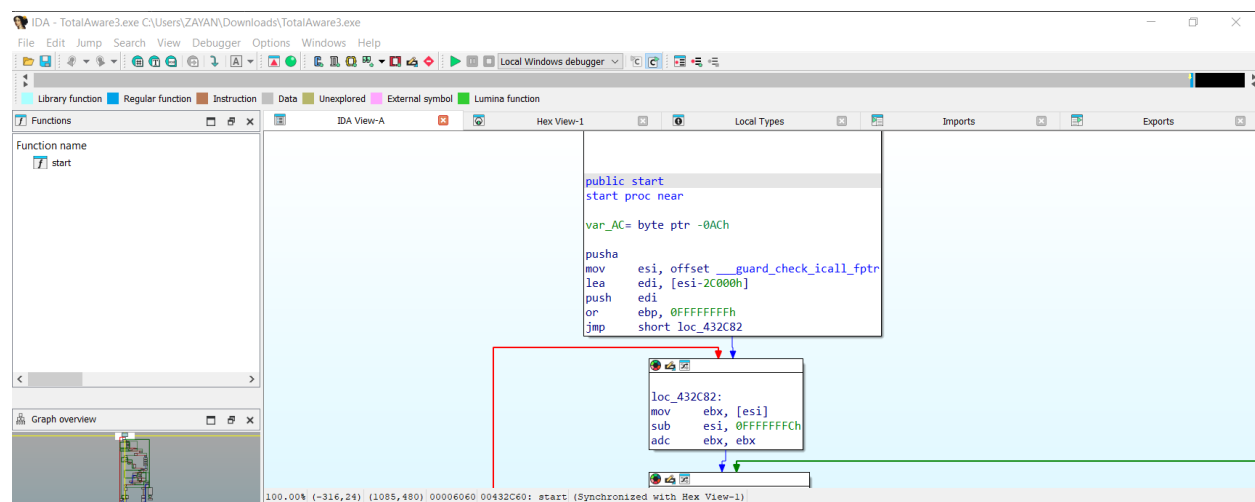
OFTs	FTs (IAT)	Hint	Name
Dword	Dword	Word	szAnsi
N/A	000332EC	0000	LoadLibraryA
N/A	000332CE	0000	ExitProcess
N/A	000332DC	0000	GetProcAddress
N/A	000332FA	0000	VirtualProtect

- Similarly, open this unpacked file, it will tell you its executable file.

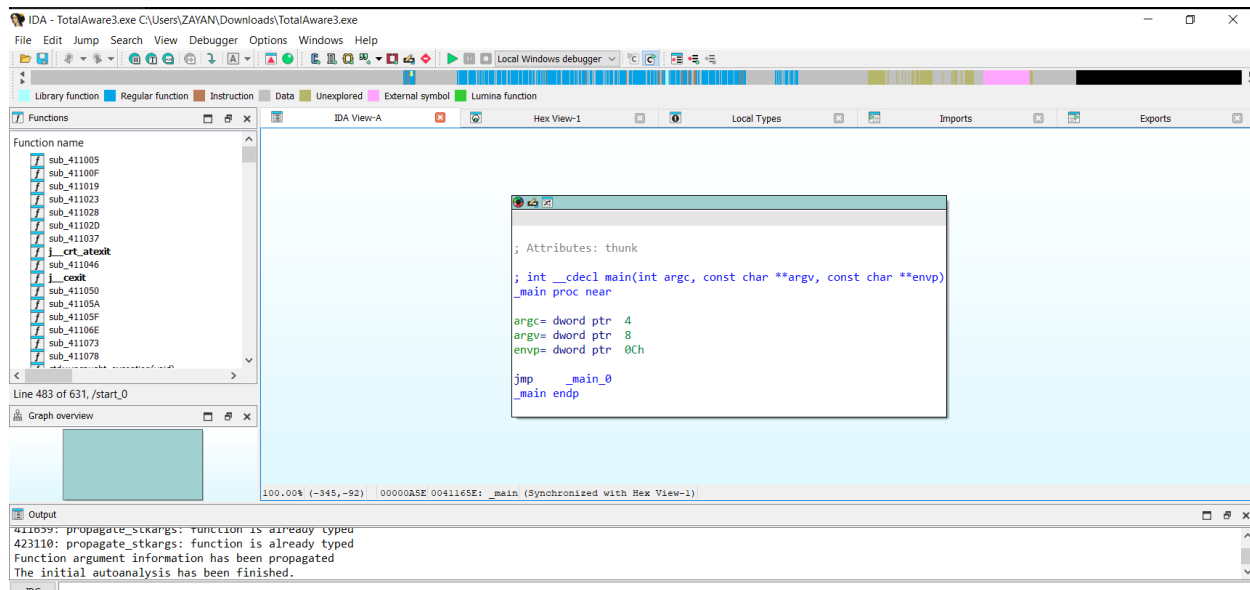
In **IDA tool**, when we open malware which will be packed, it'll show us message like some contents are not visible. It will then indicate that the malware is obfuscated.



- If file is packed, the color of bar will be black and grey.



- If it is not, bar will be of rainbow color



Static & Dynamic Malware Analysis:

Lab: Advanced Static / Dynamic Malware Analysis

- Advanced Static Analysis: “WannaCry” Malware sample
 - MD5 Hashes
 - Strings:
 - URLs
 - Universal Naming Convention (UNC) paths
 - Language pack files
 - Command Strings
 - File paths and commands
 - String Identifiers
 - Floss using CMDER
 - CAPA using CMDER
 - PE View (Virtual Size vs RAW Data Size)
 - [UniExtract](#)
 - [PEStudio](#)
- Dynamic Malware Analysis:
 - Detonation of malware samples “Ransomware.wannacry.exe” (4.1/PMAT), “Wannacry.exe” and “Wannacry-plus.exe”
 - Compare the MD5 Hashes to confirm about their uniqueness
 - Identify the detonation behaviour of the 3 x samples (provided)
 - You may use following tools:-
 - Port monitoring using TCP View
 - Process Monitor
 - FakeNet
 - Network monitoring using Wireshark
 - RegShot
 - Debugger

1. Go to the browser > Search Zoo malware > malware > binaries > Download Wanna Cry and Wanna Cry plus Ransomware

- We can also find this sample in our Flare-VM

Pmat labs > labs > 4.1 > 3rd sample is Wanna Cry Ransomware

2. Put these 3 samples in 1 folder and find their hashes and strings and compare them if they are same or different.

Remember: Default universal password for malware samples is **infected**

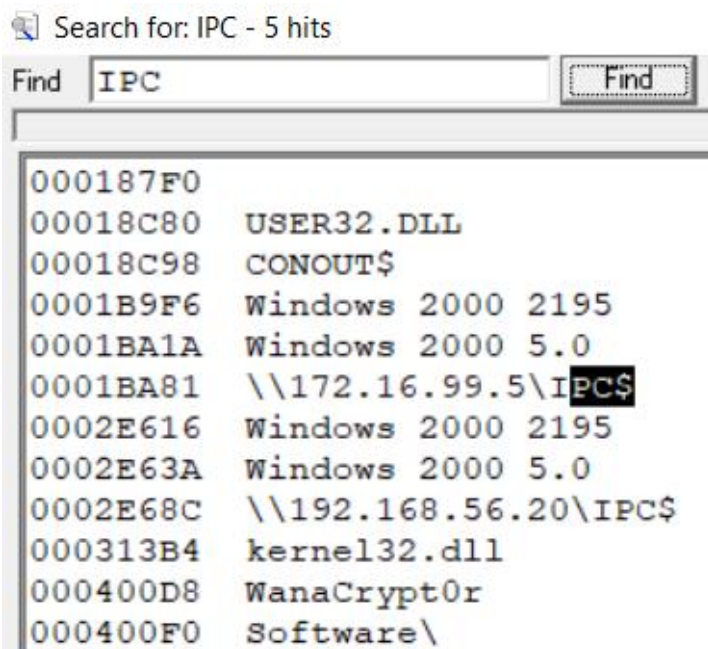
3. Find strings using **floss** in **cmdr** tool.

*In cmdr, go to the directory where floss is and type **FLOSS.exe** followed by sample path.*

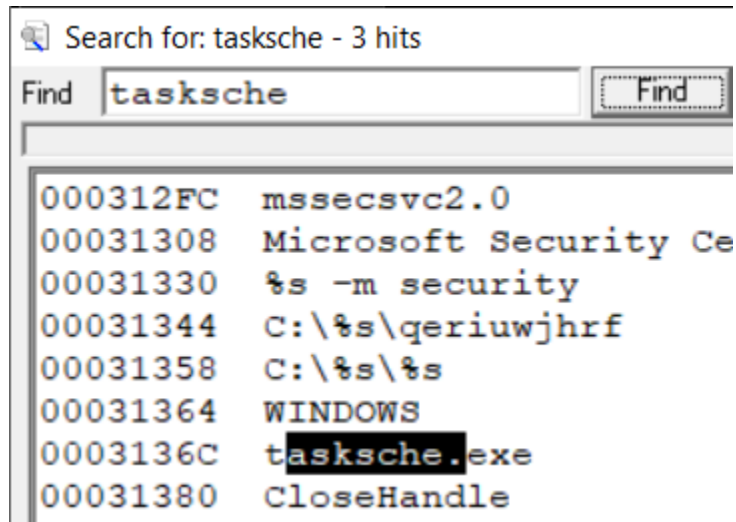
e.g: FLOSS.exe C:\users\ZAYAN\Desktop\WannaCry\Ransomware.wannacry > mydata.txt

It will find strings and save them into txt file. If you want to read the file just type **more mydata.txt**

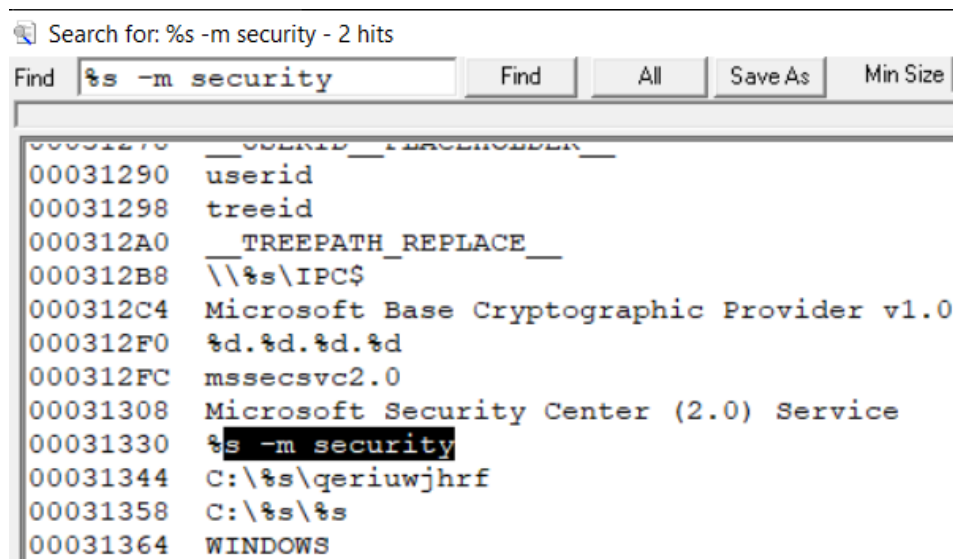
4. Find **IPC** in strings, you'll find IP. It is indicating that it is trying to access common sharing so that it can spread through sharing. Anyone who shares or take file will get infected. What's the purpose of .exe file accessing sharing resources? It indicates that it wants itself to spread.



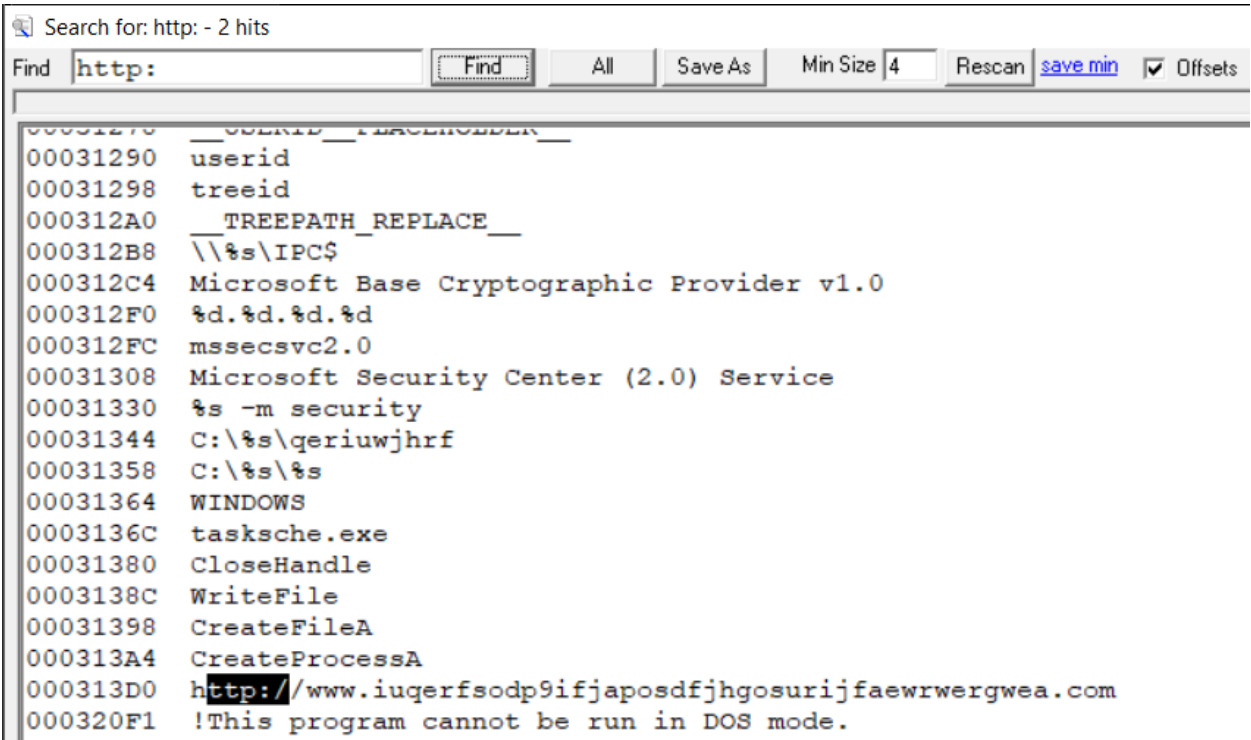
5. Find **tasksche**, this means that it will schedule tasks so it can make itself persistent.



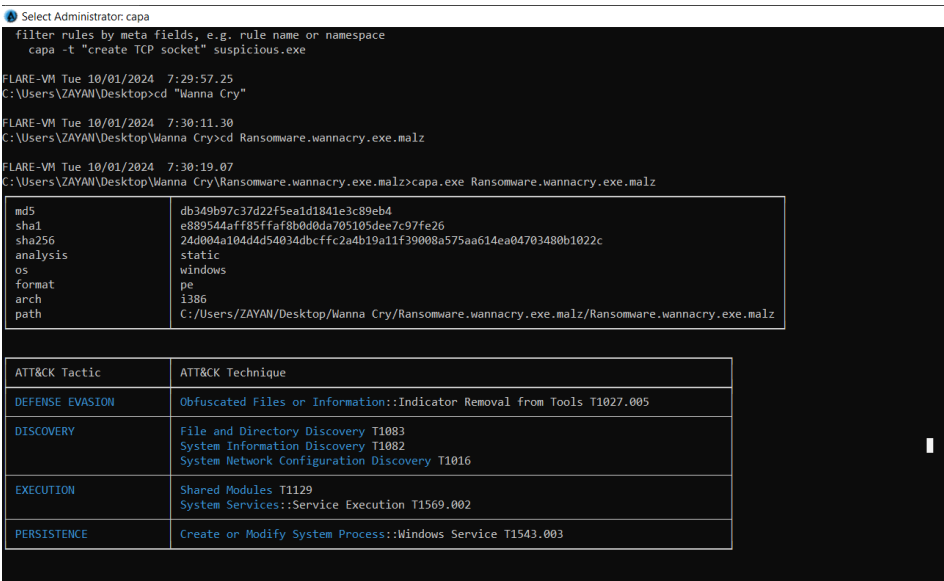
If you find %s symbol, it indicates it is placeholder like the result of execution of some command will be pasted here.



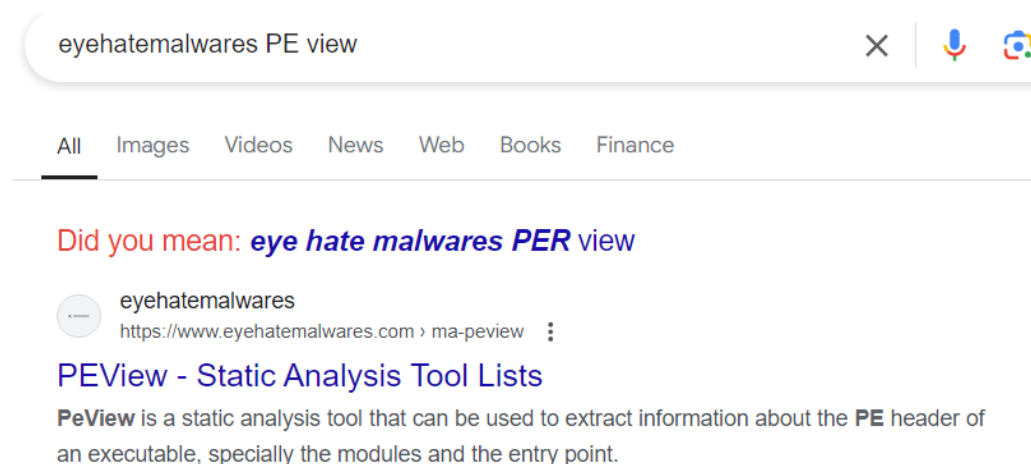
You can see unusual URL here. This surely means it is redirecting traffic to some kind of malware



CAPA:

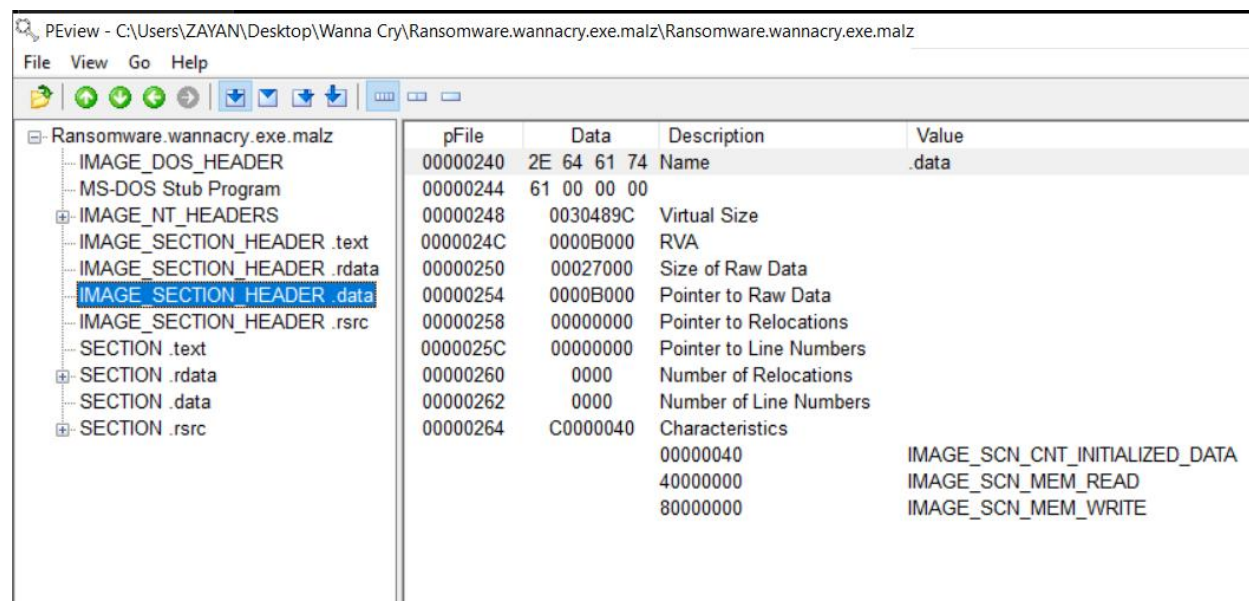


Eyehatemalwares PE view site to download **PE tool**.



1. Right click on Download Tool button > Save link as
2. Open Malware in **PE** tool

If Virtual size is more than Raw Data size, it means that file/sample is packed. When it executes, its functionality will increase.



Now we are going to do **Dynamic Analysis**:

1. Open the tool **tcpview**

TCPView - Sysinternals: www.sysinternals.com

File Edit View Process Connection Options Help

4 TCP v4 6 TCP v6 4 UDP v4 6 UDP v6 Search

Process Name	Process ID	Protocol	State	Local Address	Local Port	Remote Address	Remote Port	Create
svchost.exe	2124	UDP		127.0.0.1	1900	*		10/1/2024 8:36:5
svchost.exe	2124	UDP		150.1.7.102	1900	*		10/1/2024 8:36:5
svchost.exe	2124	UDP		192.168.5.156	1900	*		10/1/2024 8:36:5
dashHost.exe	4772	UDP		0.0.0.0	3702	*		10/1/2024 8:37:0
svchost.exe	2124	UDP		0.0.0.0	3702	*		10/1/2024 8:36:5
svchost.exe	2124	UDP		0.0.0.0	3702	*		10/1/2024 8:36:5
dashHost.exe	4772	UDP		0.0.0.0	3702	*		10/1/2024 8:37:0
svchost.exe	420	UDP		0.0.0.0	4500	*		10/1/2024 6:23:4
svchost.exe	1152	UDP		0.0.0.0	5050	*		10/1/2024 6:24:0
chrome.exe	4992	UDP		0.0.0.0	5353	*		10/1/2024 8:37:0
svchost.exe	1428	UDP		0.0.0.0	5353	*		10/1/2024 8:36:5
chrome.exe	4992	UDP		0.0.0.0	5353	*		10/1/2024 8:37:0
svchost.exe	1428	UDP		0.0.0.0	5355	*		10/1/2024 8:36:5
svchost.exe	2124	UDP		0.0.0.0	52231	*		10/1/2024 6:24:0
dashHost.exe	4772	UDP		0.0.0.0	52800	*		10/1/2024 8:37:0
svchost.exe	420	UDP		127.0.0.1	57572	*		10/1/2024 6:23:4
svchost.exe	2124	UDP		192.168.5.156	5353	*		10/1/2024 8:36:5

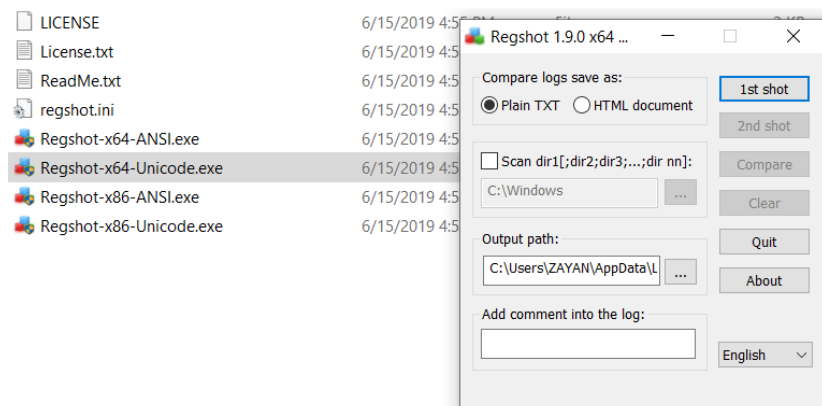
Endpoints: 75 Established: 2 Listening: 25 Time Wait: 3 Close Wait: Update: 2 sec States: (All)

Green color indicates that system is running fine.

- Run the Ransomware and look again in **tcpview**, color will change to red.

Also use **Fakenet** to check if this malware is sending http requests or trying to establish a backdoor connection.

- Open the tool **Regshot** > Click on 1st shot to take a shot



- Run the malware and then take 2nd shot > Compare the both shots and see what changes in registry files and keys, malware has made.